

A Scalable AI-Powered Platform for Smart Agriculture Using Satellite and Sensor Data

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Abstract—This paper presents a scalable and modular AI-powered platform designed for smart agriculture. The platform integrates real-time satellite data, IoT sensor readings, and machine learning models to deliver actionable insights on crop health, yield prediction, and resource optimization. By combining Google Earth Engine’s geospatial analysis with weather APIs and supervised learning techniques, the system offers an interactive and cost-effective solution for precision farming. Experimental results demonstrate improved water usage efficiency and enhanced crop condition monitoring under varied environmental scenarios.

Index Terms—Smart Agriculture, Precision Farming, Machine Learning, Google Earth Engine, NDVI, Streamlit, Resource Optimization

I. INTRODUCTION

Agriculture is undergoing a significant transformation through the adoption of data-driven technologies. Challenges such as unpredictable weather, declining soil fertility, and inefficient water usage necessitate smarter solutions. Traditional monitoring methods lack scalability and real-time responsiveness. To address this, our system uses a hybrid AI-IoT approach, merging satellite imagery, environmental sensors, and machine learning to provide farmers with predictive insights and optimization strategies.

II. SYSTEM OVERVIEW

The architecture includes four primary components:

- **Data Ingestion:** Fetching satellite imagery (e.g., Sentinel-2), local weather metrics, and field sensor values (temperature, humidity, soil moisture).
- **Preprocessing:** NDVI and other indices are computed from imagery; sensor data is cleaned and normalized.
- **ML Inference:** Trained models evaluate crop health and predict irrigation needs.
- **Visualization Layer:** A Streamlit dashboard enables user interaction and report generation.

III. VEGETATION HEALTH MODELING

Vegetation indices serve as proxies for photosynthetic activity. We compute NDVI:

$$NDVI = \frac{NIR - RED}{NIR + RED} \quad (1)$$

Here, *NIR* and *RED* are reflectance values in the near-infrared and red bands, respectively. NDVI values range from -1 to +1, with healthy vegetation typically above 0.3.

Additionally, the system computes:

$$EVI = 2.5 \times \frac{(NIR - RED)}{(NIR + 6 \times RED - 7.5 \times BLUE + 1)} \quad (2)$$

for improved sensitivity in high biomass regions.

IV. RESOURCE OPTIMIZATION ENGINE

Using data from weather APIs and soil sensors, the irrigation model estimates optimal water input I^* using:

$$I^* = f_{\theta}(T, H, M, C, W) \quad (3)$$

where T = temperature, H = humidity, M = soil moisture, C = crop type, and W = wind speed. f_{θ} is a regression model (Random Forest Regressor) trained on labeled agricultural datasets.

V. PLATFORM IMPLEMENTATION

A. Technologies Used

- Google Earth Engine for satellite data retrieval and geospatial analytics
- Streamlit for front-end UI and interactive dashboards
- Python libraries: scikit-learn, geemap, numpy, plotly
- OpenWeatherMap API for real-time weather data

B. Workflow Pipeline

- 1) User selects farm coordinates and crop type.
- 2) NDVI maps and weather trends are retrieved and analyzed.
- 3) Models infer irrigation needs and potential stress zones.
- 4) Recommendations are visualized and downloadable as reports.

VI. EXPERIMENTAL EVALUATION

A. Setup

Field data from three plots were collected over 60 days. Metrics include NDVI, rainfall, temperature, and yield.

B. Results

- NDVI-based detection achieved 88% accuracy in identifying low-growth regions.
- Water usage reduced by 24.5% through optimized irrigation alerts.
- System latency averaged below 1.8s for user requests.

VII. CONCLUSION AND FUTURE WORK

The proposed system effectively combines AI and remote sensing to enhance decision-making in agriculture. Future enhancements include pest/disease detection, multi-season yield forecasting, and farmer-specific personalization based on economic thresholds and crop calendars.

ACKNOWLEDGMENTS

We acknowledge the support of Google Earth Engine and OpenWeatherMap for their APIs and the faculty advisors at Amrita Vishwa Vidyapeetham.

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